Reputation & Feedback Intelligence Engine

Automation Documentation — TS ACADEMY Capstone Project for Group 77

Three connected n8n cloud workflows automate the full feedback lifecycle for a multi-location car-care business: a completed visit is recorded automatically (with sequential Visit and Customer IDs generated in-workflow), which triggers a feedback survey email; the customer reply is scored by AI for sentiment and severity; and every result is routed, stored and actioned in a single Airtable base with an Interfaces dashboard. All workflows are published and running.

Each stage below is one tool (n8n node). "Tool" names the component; "Work at this stage" describes exactly what it does in the flow.

System Flow

- Branch staff fill the "Mark Visit Complete" form - no IDs are typed, Visit and Customer IDs are auto-generated.
- The visit and customer records are created with auto-generated IDs (VIS001, CUS001, ...) and a feedback-request email is sent with a pre-filled survey link.
- The customer submits the public feedback form.
- The submission is posted to an intake webhook, where the AI scores sentiment, severity and confidence.
- A routing rule decides the outcome; negative feedback gets an AI-drafted reply; repeat or severe negatives escalate to the location manager by email.
- Everything lands in one Airtable base, surfaced in an Interfaces dashboard that can be filtered by location.

Workflow 01 — Visit Complete → Feedback Request

Triggered by the internal form at https://lionabiola01.app.n8n.cloud/form/mark-visit-complete.

Stage	Tool (Node)	Work at this stage
1	Form Trigger (On form submission)	Renders the internal "Mark Visit Complete" form with four fields: Customer Name, Customer Email, Location and Service. Emits them as JSON; no ID fields are collected because Visit and Customer IDs are generated automatically in this workflow.
2	Airtable - Search (Lookup All Visits)	Lists every Visit ID on file so the next sequential VIS number can be computed. Empty-safe.
3	Code (Compute Next Visit ID)	Scans the highest VIS### and emits next_visit_id (VIS001, VIS002, ...) merged with the form data.
4	Airtable - Search	Lists all customers so returning customers are

	(Lookup All Customers)	matched by name and their existing CUS### is preserved.
5	Code (Compute Customer Plan)	Finds the highest CUS### and decides next_customer_id: a new number for a brand-new customer, or the customer's existing ID when their name already exists; also surfaces the existing record id when known.
6	Airtable - Upsert (Upsert Customer)	Upserts the Customers table matching on Customer Name and writes the decided Customer ID; returns the record's id.
7	Code (Merge Customer Link)	Merges the upserted record id (customer_record_id), the assigned Customer ID and the generated Visit ID into one output item.
8	Airtable - Create (Create a record)	Writes one new row in the Visits table: completion time in Africa/Lagos, linked Customer and Location, Service, the generated Visit ID, and n8n Trigger ID = the generated VIS### (the match key the intake workflow uses later). Feedback Request Sent = false.
9	Gmail - Send (Send Feedback Request)	Emails the customer at their address. The subject names the location; the body greets them and carries a pre-filled link to the public feedback form, encoding customer_id (CUS###), customer_name, customer_email, visit_id (VIS###) and location - the customer never retypes these.
10	Airtable - Update (Mark Request Sent)	Finds the just-created visit by n8n Trigger ID and sets Feedback Request Sent = true plus the sent timestamp, closing this stage.

Workflow 02 — Customer Feedback Form

Triggered by the public form at https://lionabiola01.app.n8n.cloud/form/feedback. Identity fields are hidden and pre-filled from the email link.

Stage	Tool (Node)	Work at this stage
1	Form Trigger (Feedback Form)	Renders the customer survey. customer_id, customer_name, customer_email, visit_id and location are hidden pre-filled inputs (filled from the email link); the "How did your visit go?" textarea is the only required visible field.
2	Code (Build Intake Payload)	Reads the submitted body and normalises it into one clean object with exactly six keys (customer_id, customer_name, customer_email, visit_id, location, feedback), defaulting blanks to empty strings.
3	HTTP Request — POST (Forward to Intake)	Posts the payload as JSON to the Workbook 03 intake webhook so scoring and routing can run.

Workflow 03 — Feedback Intake, Scoring & Routing

Triggered by a POST to https://lionabiola01.app.n8n.cloud/webhook/feedback-intake (no auth, JSON body with customer_id, customer_name, customer_email, visit_id, location, feedback).

Stage	Tool (Node)	Work at this stage
1	Webhook Trigger (Webhook - Receive Feedback)	Receives the JSON body (customer_id, customer_name, customer_email, visit_id, location, feedback) from the feedback form or any manual client and answers immediately so the caller gets a fast response while the pipeline continues.
2	Code - Validate (Validate Input)	Checks the six fields, records which are missing, and generates the anti-duplicate token (FB-<timestamp>-<random>).
3	Airtable - Search + Code (Lookup All Feedback / Compute Next Feedback ID)	Reads every existing Feedback ID and emits the next sequential FID###, used by both the routing and the human-review branches. Empty-safe.
4	If - condition (Input Valid?)	Branches: complete submissions go to history and scoring; incomplete ones go to the human-review fallback.
5	Set + Airtable - Create (Flag Missing Data / Save Invalid)	Tags Route = Human Review and Status = Needs Human Review and stores the submission in the Feedback table with its FID###, so weak or blank submissions are never scored and are queued for a person.
6	Airtable - Search (Get Customer History)	Reads every prior Feedback row for this customer name, so the engine can detect repeat complaints and duplicates.
7	Code - History (Summarize History)	Computes repeat_negative (any earlier Negative feedback), prior_count, and is_duplicate (same feedback ID already stored).
8	If - condition (Is Duplicate?)	Routes true duplicates to the skip branch; genuine reviews continue to AI scoring.
9	Set (Skip Duplicate)	Writes Status = Skipped - Duplicate so re-posted submissions are never double-scored or double-alerted.
10	OpenAI - Chat model (AI - Analyse Feedback)	GPT-4o-mini with the scoring rubric and structured outputs; it must return exactly the JSON schema sentiment, sentiment_score, severity, severity_score, confidence_score and reason - at low temperature (0.2) for consistency.
11	Code - Validate (Validate AI Output)	Parses and validates the model reply, converts scores to numbers, and sets ai_valid = false if anything is missing or unparseable, so bad AI output can never corrupt routing.
12	Code - Route (Determine Route)	Applies the routing matrix, picks the location manager email, and sets needs_draft, status, route and alert_reason.
13	Airtable - Search (Lookup Visit)	Finds the Visit whose n8n Trigger ID equals the submitted visit_id, so the feedback attaches to the right visit. Fail-soft when the visit is missing.

14	Airtable - Create (Save Feedback)	Stores the fully scored record in the Feedback table (text, scores, sentiment, severity, confidence, route, status, AI reason) with Feedback ID FID###, linking Customer, Location and the matched Visit.
15	Airtable - Update (Update Visit Flag)	Matches the visit by n8n Trigger ID and sets Feedback Collection Triggered = true; the Visit's Feedback link fills in automatically.
16	If - condition (Needs AI Draft?)	Only negative feedback needs a draft reply; positive and neutral feedback skip the drafting branch entirely.
17	OpenAI - Chat model (Generate AI Response)	GPT-4o-mini drafts a personalised reply naming the customer and referencing their specific issue, following the policy (no liability, no promises, no claims of resolution; acknowledges repeat complaints).
18	Airtable - Search + Code (Lookup All Responses / Compute Next Response ID)	Emits the next sequential Response ID RES###. Empty-safe.
19	Airtable - Create (Save Response)	Stores the AI draft in the Responses table with Response ID RES### and Human Review Status = Pending Review, linked to the feedback record, so a human approves before anything is sent.
20	If - condition (Is Manager Alert?)	Checks whether the route is Manager Alert.
21	Airtable - Search + Code (Lookup All Alerts / Compute Next Alert ID)	Emits the next sequential Alert ID ALT###. Empty-safe.
22	Airtable - Create (Create Alert)	Creates the escalation record in the Alerts table with Alert ID ALT###: the draft as Alert Message, the reason, severity score, linked feedback and Alert Status = Open.
23	Airtable - Search + Gmail (Lookup Manager Email / Send Manager Alert Email)	Looks up the location's Manager Email from the Locations table and emails the manager with customer, scores, reason, original feedback and the AI-drafted reply for review. Fail-soft so an email problem never kills the pipeline.

Routing Rules

Condition	**Route**	**Status**
Missing data, or AI invalid / confidence < 75	Human Review	Needs Human Review
Positive	Ready to Post	Pending Human Review
Negative → customer has complained before	Manager Alert	Escalated - Repeat Negative
Negative → severity score ≥ 70	Manager Alert	Escalated - Severe
Negative (first time, moderate)	Private Queue	Pending Human Review
Neutral	Private Queue	Pending Human Review

Data (single Airtable base)

- Visits — each completed visit; auto-generated VIS###, n8n Trigger ID (the match key), feedback-request flags and the Feedback Collection Triggered flag.
- Feedback — every submission with AI scores, sentiment, severity, confidence, route, status, reason and linked visit/customer/location.
- Responses — AI-drafted reply for each negative item, pending human review.
- Alerts — escalation records for repeat or severe negatives.
- Customers & Locations — records referenced by the link fields above.

ID Scheme

• Visits — VIS### (e.g. VIS001) is generated in Workflow 01 from the highest existing number, stored both as the Visit ID and as the n8n Trigger ID match key.

• Customers — CUS### is generated the same way; a returning customer (same name) keeps their existing CUS### instead of getting a new number.

• Feedback — FID### is generated in Workflow 03 for both the scored branch and the human-review branch.

• Responses — RES### is generated whenever a negative draft reply is created.

• Alerts — ALT### is generated whenever an escalation record is created.

• Primary ID fields are plain text (converted from Auto Number) so the workflows can read and compute them; the old formula-based ID columns have been removed.

Dashboard (Airtable Interfaces)

An Airtable Interfaces app built on the same base gives management the single-picture view required by the brief:

• Overview — charts for feedback volume by location and the sentiment split.

• Every feedback — a grid of all submissions with Customer, Location, Sentiment Score, Severity Score, Route and Status.

• Ready to Post queue — the copy-and-paste list of positive reviews awaiting publication.

• Negative feedback + drafts — each complaint beside its AI-drafted reply and the Sent checkbox, so the team sees whether a reply has gone out.

• Open alerts — an escalation board with editable Alert Status (Open / Investigating / Resolved / Archived) so managers close out handled alerts.

• Location filter — an interactive record picker filters every panel to a single branch.

(Paste dashboard screenshot here.)

How to Run / Test

- Live: staff submit the "Mark Visit Complete" form (IDs are auto-generated); the customer then completes the pre-filled feedback form. The pipeline runs automatically.

Configuration Notes

- Credentials: Airtable (OAuth, main base) and Gmail (OAuth) are connected; AI uses OpenAI GPT-4o-mini with a structured JSON schema and low temperature (0.2).
- Manager emails are looked up from the Locations table (Manager Email field); if a location has no email set, the workflow falls back to lionabiola@gmail.com. Change the Locations table to control who receives escalation alerts.

Sequential IDs (VIS/CUS/FID/RES/ALT) use a read-then-write pattern that is empty-table safe (the lookup searches are configured to still emit when a table is empty), so the very first submission on an empty base works.

The n8n Feedback ID token (FB-<timestamp>-<random>) is the anti-duplicate guard; the branded sequential IDs are what appear in the Airtable base and dashboard.

- Time zone for visit timestamps is Africa/Lagos.

Submission Materials

• Demo video (Loom or screen recording): walk through what was built and run the three paths live - a positive case, a first-time negative case, and a repeat-negative case.

• Screenshots: one of each workflow canvas (Workflow 01, 02 and 03) and of the dashboard.

• One shared Google Drive folder set to "Anyone with the link", containing the screenshots (and the video file if it is not on Loom).

• Every group member submits the same links (video + Drive folder) individually via the capstone Google Form.